TECHNOLOGY THAT MATTERS

THE PRACTICAL APPLICATION OF SOA, EJB, WEB SERVICES AND MORE

white paper

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INTRODUCTION: KNOW WHAT MATTERS

Service Oriented Architecture (SOA) is fundamentally changing the way in which enterprises deploy information technology (IT) to support business operations.

Historically, IT departments have held an application-centric view of the world. Most of their budgets have been allocated to the purchase, deployment and maintenance of individual applications. In more recent times, a growing percentage of the budget has been allocated to integration projects, in an attempt to deliver broader and more seamless support for business processes. However, these projects have often fallen short of their promise due to the inherent rigidity of hard-wired integration: any change to a single application must be propagated across the entire integration, creating an expensive and change-averse infrastructure.

SOA enables IT departments to make the transition from an application-centric view of the world to a process-centric one. For the first time ever, IT departments have the freedom to combine business services from multiple applications to deliver true end-to-end support for business processes. And, because the integration mechanism—usually Web services—enables loosely coupled integration, IT can upgrade or change applications without impacting other applications in the SOA.

Infor has responded to this fundamental shift in the technology landscape by making Datastream 7i fully SOA-enabled, with a robust Web services foundation. This enables customers to readily expose Datastream 7i functionality to other applications and interfaces, thus radically expanding its value and return-on-investment.

Infor has always built its products based on a simple guiding principal: technology matters. Technology that helps customers achieve their business objectives matters. SOA matters, because it unlocks enormous potential for customers to improve efficiency and reduce operations and IT costs.

With virtually every application vendor jumping on the SOA bandwagon today, this white paper clarifies what it means to be SOA-enabled by discussing technologies that truly matter, including Web services and Enterprise JavaBeans, along with practical examples of how they are being used today by Infor customers.

PRACTICAL ARCHITECTURE AND DESIGN

Service Oriented Architecture

IT's core mission is to use information technology to help the enterprise achieve its objectives with maximum efficiency. In a perfect world, IT systems would provide seamless support across business processes and have the inherent flexibility to adapt to changing conditions. For example, a perfect manufacturing application would support every step of the manufacturing process with robust functionality—from materials procurement and warehousing, to execution, maintenance, quality, time and labor and shipping logistics. The application would be able to adapt to changing market demands, unexpected supply chain exceptions, new regulations, etc., while also providing complete visibility from start to finish on the state of the process.

Historically, achieving this level of integration has required tradeoffs. Deploying best-of-breed applications delivers the required functionality, but also results in application "silos" that are difficult to integrate. Deploying the offerings of application suite vendors delivers the required integration, but the functionality of the individual applications is lacking and deployments are legendarily complex and costly.

Enter Service Oriented Architecture. SOA is the emerging IT architecture in which application functionality is exposed as "business services" across a standards-based communications infrastructure. Web services are the key enabling technology to SOA, providing a common framework for applications to interoperate. This enables companies to tear down application silos and replace them with libraries of business services that can be accessed as needed to support business processes.

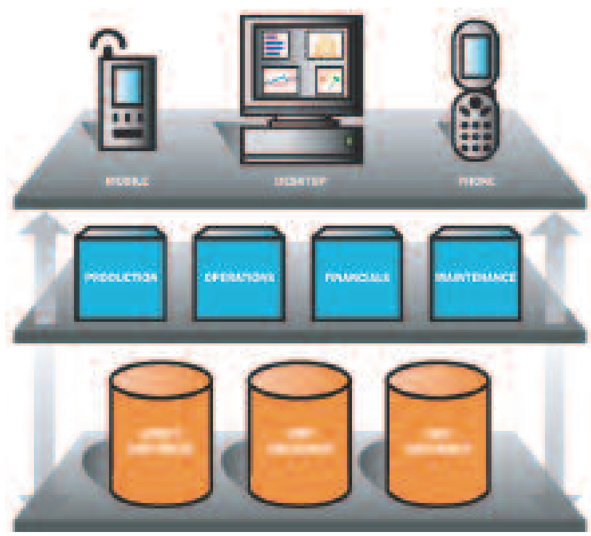
Within an SOA, the traditional standalone application paradigm is replaced by a process-centric view of IT systems, in which business services are called upon to provide seamless end-to-end support for business processes. However, SOA is only feasible if applications are truly SOA-ready.

Datastream 7i is the only product in its class that is fully architected for SOA by enabling more than 1,500 Web services integration points. This provides one of the industry's broadest solution sets for integrating with complementary applications such as CRM, GIS, ERP and SCADA.

Customers in a variety of industries—including automotive manufacturing, oil & gas companies, transportation agencies, pharmaceutical and life science, food & beverage, and public works and water/wastewater treatment—are directly benefiting from this architecture today. Additionally, SOA-readiness enables Datastream 7i to deliver unprecedented support for managing geographically distributed assets by enabling advanced capabilities including:

- **Full Mobile Support** One of the most powerful aspects of SOA is its ability to extend the reach of application functionality to the most remote edges of the enterprise (see Figure 1). And the ultimate edge is the mobile application. The Datastream 7i Web services foundation enables customers to deploy on virtually any handheld device—from Tablet PCs to handheld computers and even cellular telephones. This frees customers from the limitations of custom-built hard-wired mobile applications by eliminating the inherent deployment, maintenance and integration problems.
- **Integration to GIS Applications** The Datastream 7i Web services foundation enables full integration with leading GIS applications, such as those from ESRI. This enables GIS applications to be accessed directly from Datastream 7i, and vice versa, thereby radically simplifying and streamlining the process of locating and maintaining geographically dispersed assets.

Figure 1: SOA Across the Enterprise



The reach of application functionality extends to the most remote edges of the enterprise.

SOA in Practice

SOA can enable the seamless integration of multiple critical business applications, such as an EAM system with a production system. One example includes a leading manufacturer of frozen food products that needed to monitor its mission-critical refrigeration units. Because of the underlying SOA architecture, this manufacturer was able to easily integrate its refrigeration alarm system with its EAM system. The result was the ability to monitor and track exception alarms coming from the refrigeration management system and automatically create work orders for maintenance technicians to immediately respond to these exception alarms. The SOA platform has allowed this manufacturer to easily integrate two disparate systems to detect problems coming from the refrigeration management system as they occur, which has significantly reduced the potential spoilage of its most profitable product lines and prevented any disruption to the supply chain.

Another practical application includes a large municipality using SOA to integrate Datastream 7i with CRM, GIS and financial systems, to improve responsiveness and reduce costs. Citizens can phone into a call center when there is a town-related problem (for example, a leaking fire hydrant). When that happens, a Web services call from the CRM system triggers a work order in Datastream 7i. The worker can then locate the hydrant in a GIS application, map out the shortest route to it, and then click on the hydrant in the GIS application, invoking another Web services call into Datastream 7i that pulls up the entire asset history.

Meanwhile, if other citizens call about the hydrant, call center personnel can access repair status information from Datastream 7i through their familiar customer relationship management (CRM) system, again via Web services. When work is completed, Datastream 7i closes out the work order and sends time and labor information into the financial application for cost accounting, while also sending a Web services call to the CRM system that queues up phone calls to citizens who phoned in about the hydrant, so they can be notified that the problem has been taken care of.

These are classic SOA success stories where formerly discrete applications—EAM and alarm systems, or EAM, CRM, GIS and financials—work in seamless cooperation to fully support a business process: be it monitoring refrigeration units or repairing a fire hydrant. And because of the loosely coupled integration in SOA, any change to one of those applications (upgrade, change in vendor, etc.) will not impact the others or the overall integration.

SOA in this case works as advertised because all of the systems involved have the inherent foundation in place to generate and consume Web services in real time. This is an important distinction because not all "Web services-enabled" products provide this caliber of support. Some only support "asynchronous" Web services, which means they can only perform batch transfers of data. While this can be useful for certain functions, it does not enable real-time support for business processes such as the one described above.

The Bottom Line:

- Web Services are critical to the SOA because they enable loosely coupled interoperability between applications.
- In order to support business processes SOA-enabled applications must provide not only classical batch imports, but also real-time support for consumption and production of Web Services.
- Datastream 7i is fully SOA-enabled with more than 1,500 Web Services integration points.
- Multiple large enterprises are successfully using Datastream 7i in SOAs today.

Enterprise JavaBeans

Enterprise JavaBeans (EJBs) are generally considered a required component of a robust Java 2 Enterprise Edition (J2EE) architecture. Indeed, it does not make sense to invest in J2EE application servers such as WebSphere and WebLogic if one does not plan to manage EJBs. Without EJBs, most of the advanced features in these products—such as clustering, pooling, caching, load balancing and failover—are irrelevant.

The major benefit of EJBs is they force the separation of presentation and business logic in applications. This radically improves application scalability and flexibility, because business logic is centrally located on the server and accessible to any number of user interfaces. Changes made to the business logic do not impact the client interface, and vice versa.

EJBs are ideally suited to SOA because they make it relatively easy to expose specific application functionality as business services, via Web services. Some J2EE applications do not have this EJB tier and instead rely on JavaBeans as their component architecture. In these cases, there is an intermingling of business logic and presentation, which impairs the application's ability to participate in SOA. According to Gartner, "To establish services, the business logic must be dissected, then 'isolated' from the user interface/forms."[1]

Mobile applications may be the best illustration of the benefits of EJB. Because EJBs encapsulate business logic and create the necessary isolation from the presentation layer, mobile applications can be deployed on virtually any mobile platform, and changes to the business logic do not impact the mobile clients, and vice versa. With JavaBeans, business logic resides on the client and the server, which means mobile applications must be purpose-built for a specific platform, and any change to the business logic requires changes to both the client and server.

As Sun Microsystems explains in its publication, Java Server Programming, "The JavaBeans architecture is meant to provide a format for general-purpose components, whereas the Enterprise JavaBeans architecture provides a format for highly specialized business logic components." Since SOA requires encapsulating and exposing business logic for consumption by other applications, an EJB layer is a requirement for SOA-ready applications.

Datastream 7i boasts a true J2EE architecture with an EJB layer, enabling customers to gain all of the scalability and flexibility benefits that come with EJBs. EJBs also "future proof" Datastream 7i from changing business requirements. For example, EJBs can work with any number of clients—including Java Server Pages (JSP), Java clients, CORBA and XML—which protects customers from being locked in to "HTML-only" client architectures. This also means that EJBs can be accessed from virtually any interface, even those of other applications.

Perhaps most important of all, though, is that EJB enables Datastream 7i to interoperate with components that are not part of the J2EE stack, including .Net components. This is a critical capability since, according to Forrester Research, nearly one-third of enterprise applications are written in .Net programming languages.[2] This means that J2EE products lacking an EJB layer are simply not practical in the era of SOA, where it is critical for application components to interoperate with non-HTML centric applications.

1 Gartner, "SOA Will Demand Re-Engineering of Business Applications," Y. Genovese, S. Hayward, J. Comport, Oct. 8, 2004.

2 Forrester, "Of Strategic Languages, Java's Adoption is Highest," July 12, 2005

EJBs in Practice

Hundreds of Infor customers are reaping rewards from the Datastream 7i EJB layer because it enables them to extend the system in innovative new ways.

For example, a large government agency selected Datastream 7i due in large part to its EJB layer. The agency needed its application to be "508 compliant," the government mandate for software to be designed for use by the blind and handicapped. Because Datastream 7i has an EJB layer and can work with any client, it was relatively trivial to develop special interfaces for these users. And, the business logic can be reused without impacting the client, which radically simplifies and reduces the costs of ongoing application ownership.

Another customer wanted to access Datastream 7i work orders directly through a GIS application. Again, because of the Datastream 7i EJB layer and support for Web services, the application could readily adapt to this customer requirement.

The Bottom Line:

- EJB is a critical layer for applications participating in SOAs because it encapsulates business logic for use by any number of clients.
- Many J2EE applications lack an EJB layer and use simple JavaBeans as their component architecture. This intermingling of business logic and presentation makes it difficult to extend applications to multiple clients and also make reuse more expensive.
- EJB also enables J2EE applications to interoperate with other component architectures, like .Net.
- Datastream 7i is a complete J2EE application with an EJB layer.

PRACTICAL INTEGRATION

Web Services

Web services are usually inextricably linked with SOA. But in reality, Web services are simply an enabling framework for SOA, rather than a required component for one. Web services themselves can be used without having SOA in place. Likewise, companies can deploy SOAs that do not utilize Web services. (Any distributed component architecture is technically an SOA. CORBA, for example, is an SOA. Web services are considered key to making SOAs practical, because they provide a relatively easy-to-implement mechanism for enabling application interoperability.)

Many Infor customers have deployed Datastream 7i as part of their strategic SOA projects. More still, however, are simply using Web services to extend Datastream 7i functionality in innovative new ways.

Web services are based on an industry-standard protocol stack that includes individual protocols for discovery, description and remote service calls, as well as HTTP and TCP/IP (see Figure 2, at right). The protocols in widest use today include WSDL (for discovery) and SOAP (for calling business services).

Figure 2: Web Services Protocols

WEB SERVICES
UDDI (DISCOVERY)
SOAP (REMOTE SERVICE CALL)
HTTP (TRANSPORT APPLICATION PROTOCOL)
TCP/IP (TRANSPORT PROTOCOL)

Protocols in widest use today include WSDL (for discovery) and SOAP (for calling business services).

The Web services implementation in Datastream 7i enables customers to use SOAP calls to access functionality from other applications from within Datastream 7i, and vice versa. It also enables unlimited user interface options on virtually any fixed or mobile computing device. For example, customers can extend Datastream 7i to virtually any mobile computing platform or application, and more fully support critical business processes.

Web Services in Practice

Web services are in use today for a broad range of purposes, from rudimentary batch data transfer to more advanced application extension and SOA integration. Hundreds of Datastream 7i customers are using its Web services functionality to extend the solution in new, innovative ways. For example:

- A large pharmaceutical company had standardized on open source browsers, rather than Internet Explorer. Software products supporting only "pure HTML" could not meet the company's needs. But Datastream 7i could because its Web services support made creating an interface for the open source browser a trivial task.

- One of the world's largest diversified product companies wanted to deploy a customized inspection application on Tablet PCs and have it integrate with Datastream 7i. Through Web services and Enterprise JavaBeans, Datastream 7i enabled the application to be developed and deployed in a matter of weeks. And, because Web services-based integration is loosely coupled, application components are isolated from one another. This means that the customer can change its inspection application, and Infor can upgrade Datastream 7i, without having to worry about breaking the integration.
- A global facilities services company wanted to integrate its call center application with Datastream 7i to streamline service response time. The company used the Datastream 7i Web services foundation and Databridge integration capabilities to create an innovative system that enables call center personnel to create a work order in Datastream 7i through their familiar call center application and then, with the push of a button, page the technician and transmit the work order as an XML document. This enables technicians to download work orders in the field so they can move from job to job with maximum efficiency.

These are just three examples of how Web services are being used today to streamline work processes, accommodate existing IT infrastructure and enable new kinds of integrated process-centric solutions.

The Bottom Line:

- Web Services enable customers to extend Datastream 7i in an infinite variety of ways.
- Web Services eliminate the problems associated with proprietary platforms and existing IT infrastructure, by enabling interfaces to be deployed on virtually any platform.
- Web Services enable customers, for the first time, to define IT systems according to business processes, rather than having processes defined by applications.

PRACTICAL PERFORMANCE MANAGEMENT

Data Warehouse, Analytics and Reporting

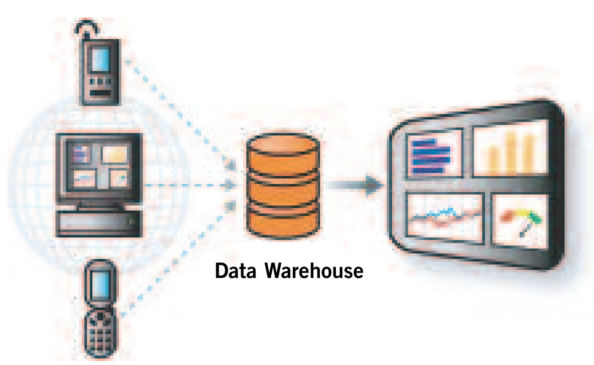
In multi-site distributed deployments, the practical benefits of a Web-architected asset management solution are clear: centralized server deployment and thin clients reduce cost of ownership, and application standardization enables work-process standardization, which facilitates the global adoption of best practices.

However, perhaps the greatest benefit to these kinds of deployments is data. More specifically, the ability to aggregate asset data from across the enterprise in a data warehouse, allows the data to be analyzed and manipulated to provide operational visibility and business intelligence.

Datastream 7i is built on the concept of Asset Performance Management—the methodology of collecting, analyzing and presenting data in such a way that customers can identify opportunities for operational improvement and make better, more informed business decisions.

The key enabler to this methodology is the Datastream 7i data warehouse (see figure 3), Datastream 7i Analytics and the Advanced Reporting module. By aggregating asset data in a single warehouse, customers can apply analytics or reporting mechanisms to the data. This allows them to identify trends and anomalies, leading to opportunities to reduce costs and improve operations. Additionally, customers can use Web services to present Key Performance Indicators and other critical metrics via executive dashboards. This provides critical "at a glance" information that executives can use to make better-informed business decisions.

Figure 3: Data Warehouse, Analytics and Reporting



The data warehouse, with analytics and reporting, enables better reporting.

Data Warehouse, Analytics and Reporting in Practice

A major automotive manufacturing service provider deployed Datastream 7i to manage its service operations worldwide. As part of the deployment, the company implemented a data warehouse for all of its asset data so it could analyze it and compare operational efficiency across plants, identify under-performing assets and determine the root causes of problems.

The company is using the Datastream 7i Advanced Reporting module, powered by Cognos ReportNet, to generate advanced graphical reports as well as a broad range of business reports, all through a single Web-based reporting solution. The company is also using Datastream 7i Analytics to perform what-if scenarios and to identify trends and anomalies in operations.

Together, this powerful combination of data warehouse, analytics and reporting enables the company to analyze asset data in new ways so business decision-makers can gain much greater visibility into the costs of operations and the performance of individual assets. For example, if a piece of equipment in Mexico City is breaking down more often than an identical piece of equipment in Detroit, the company can identify the processes used to maintain the equipment in Detroit and transfer them to Mexico City to solve the problem.

With this data warehouse, the company can:

- Detect and stop abuse or improper use of equipment;
- Run utilization reports to establish benchmarks against which future operations can be measured;
- Automatically calculate cost savings through "what if" scenarios; and
- Analyze historic data to predict future job requirements (for example, determining staffing requirements based on similar jobs performed in the past).

This is just one example of how the practical implementation of an asset data warehouse, analytics and reporting enables companies to gain complete visibility into operations and radically streamline business processes.

CONCLUSION

A traditional business application is nothing more than a software vendor's opinion of how a business process should be performed. To date, this dynamic has prevented IT departments from delivering maximum value to the enterprise because more often than not vendor opinions and real-world processes do not match. As a result, IT systems have largely failed to support business processes; rather, they have only automated portions of processes.

SOA is changing this by enabling IT departments to take a process-centric approach to architecture. This is also changing the way in which enterprises evaluate IT vendors. Those that can fully support the migration to SOA stand to thrive in this new world. Those that cannot will struggle.

By implementing technologies that matter—EJB, Web services, data warehouse, analytics and reporting—Datastream 7i is ideally suited to SOA deployment. More importantly, Infor's customers have all of the technology and capabilities required to take full advantage of the SOA future.

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